

Forecasting Wildfire PM2.5 with Temporal Convolutional Networks: A Comparison to Classical Time Series Models

Eric Huang, Zoe Zhou, Jeffrey Choi

December 19, 2025

1 Introduction

Wildfire smoke drives extreme short-term air pollution events in California, causing sudden PM_{2.5} spikes that differ fundamentally from typical background dynamics. These events exhibit rapid regime shifts, strong nonlinearity, and long temporal dependence, violating the linearity and stationarity assumptions underlying classical time series models. Consequently, traditional autoregressive approaches may respond slowly to sharp concentration increases, systematically underpredicting extreme pollution episodes critical for public health forecasting.

ARIMA and SARIMA models remain standard baselines for environmental forecasting due to their interpretability and computational efficiency. However, their fixed linear structure limits adaptability to external shocks like wildfire intrusions. Modern deep learning models have shown improved performance in capturing nonlinear temporal patterns, though many rely on complex spatial architectures or heavy computation. This motivates investigation of lightweight alternatives that balance flexibility with practicality.

We evaluate whether Temporal Convolutional Networks (TCN), augmented with wildfire proximity features, can outperform classical ARIMA baselines for short-term PM_{2.5} forecasting. Our contributions are: (1) a unified dataset aligning hourly PM_{2.5} observations with NOAA HMS wildfire data across 16 California sites, (2) ARIMA-based forecasting baselines, and (3) controlled comparison of classical statistical and lightweight deep learning approaches under matched experimental conditions.

2 Related Works

Early air quality forecasting has relied on statistical time series models such as ARIMA and SARIMA due to their interpretability and well-understood theoretical properties. However, environmental time series often exhibit nonstationary behavior and abrupt

spikes, limiting the effectiveness of purely linear approaches in extreme pollution settings. [Kontopoulou et al. \[2023\]](#) highlight the growing success of deep learning models for capturing nonlinear dynamics and rapid regime changes. Neural architectures including recurrent, convolutional, and transformer-based methods have shown improved performance over classical baselines by learning flexible temporal representations without requiring explicit stationarity assumptions.

[Yu et al. \[2023\]](#) propose a spatiotemporal architecture integrating spatial dependencies across monitoring stations with strong predictive performance. While effective, such approaches are computationally intensive, rely on dense spatial coverage, and may obscure the contribution of local temporal dynamics. Despite growing interest in wildfire-aware modeling, relatively few studies conduct controlled comparisons between classical statistical baselines and lightweight deep learning models under matched experimental conditions. This gap motivates the present work.

3 Data and Preprocessing

3.1 Data Collection

Hourly PM_{2.5} measurements were obtained from the United States Environmental Protection Agency (EPA) air quality monitoring network at 16 monitoring sites across California. The hourly resolution captures rapid pollution onset and decay of pollution events, which are often missed by coarser temporal aggregations. Wildfire activity information was obtained from the NOAA Hazard Mapping System (HMS) [\[Rolph et al., 2009\]](#), which provides manually curated smoke polygons and fire alerts derived from satellite observations.

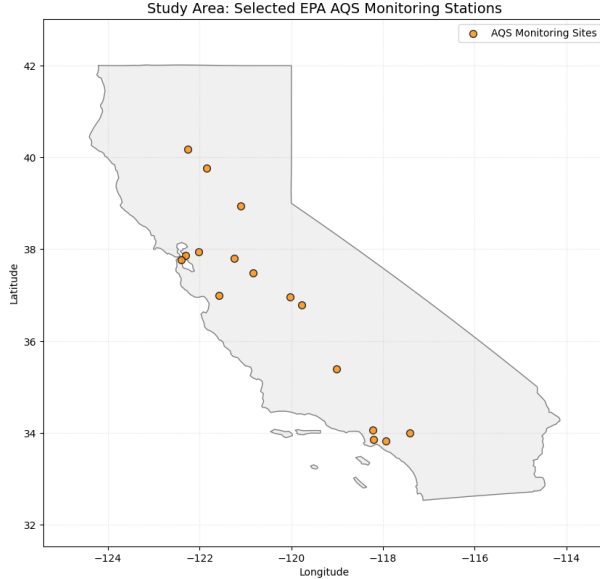


Figure 1: Geographic distribution of 16 EPA AQS monitoring sites across California.

3.2 Preprocessing and Feature Engineering

PM_{2.5} time series were sorted by site and timestamp. Missing values (≤ 8 -hour gaps) were imputed via linear interpolation; longer gaps were excluded. One site with excessive missing data ($>20\%$) was removed. Stationarity diagnostics (ADF, KPSS) revealed strong temporal dependence and nonstationarity during wildfire periods, motivating both differencing in ARIMA models and nonlinear approaches.

Wildfire proximity features were engineered by computing the distance from each monitoring site to the nearest active smoke polygon and fire event at each hour. Fire alert detections were spatially clustered and temporally aggregated into discrete fire episodes to reduce redundancy. Meteorological variables (wind speed, wind direction) were temporally aligned and forward-filled for short gaps.

3.3 Wildfire and Smoke Data

Wildfire activity information was derived from the NOAA Hazard Mapping System (HMS), which provides manually curated smoke polygons and fire alerts based on satellite observations. Each polygon is associated with a qualitative severity label indicating smoke density. To incorporate wildfire influence into the forecasting models, we engineered proximity-based features that quantify the distance between each monitoring site and nearby smoke polygons at each hour. These features provide a parsimonious representation of wildfire impact without requiring

explicit spatial modeling.

4 Methods

4.1 ARIMA-based Models

Autoregressive Integrated Moving Average (ARIMA) models were used as linear baselines for PM_{2.5} forecasting [Box et al., 2015]. Based on results from stationarity diagnostics and autocorrelation structure, we adopted a low-order ARIMA specification with first differencing to stabilize temporal dependence, ARIMA(2,1,2).

We fit ARIMA models independently for each monitoring site with an integer-indexed PM_{2.5} time series, and used ARIMA(2,1,2) specification across all 16 sites to ensure consistency and comparability. Moreover, to incorporate seasonal pattern, a 24-hour pattern, we additionally applied a SARIMA specification. The same seasonal structure was applied across sites to enable a clean comparison between seasonal and non-seasonal linear models.

We used maximum likelihood estimation for model fitting. For each site, the time series was split chronologically, with the first 80% of observations used for training and the remaining 20% for evaluation, preserving the temporal ordering of the data. Forecasts were generated over the full test horizon, and assessed predictive accuracy with mean absolute error (MAE) and root mean squared error (RMSE).

To incorporate external wildfire and meteorological information, we further extended the seasonal model to a SARIMAX formulation by including proximity-based wildfire features and local weather covariates as exogenous regressors. Exogenous variables were temporally aligned with the PM_{2.5} series and imputed using site-level means.

4.2 Temporal Convolutional Networks (TCN)

Temporal Convolutional Networks (TCN) are a class of convolutional architectures designed for sequence modeling tasks. Unlike recurrent models, TCNs process entire sequences using stacked causal convolutional layers with exponentially increasing dilation rates, enabling efficient capture of long-range temporal dependencies without recurrence.

We adopted a three-layer TCN architecture with dilations (1, 2, 4), allowing the network to aggregate information over a receptive field spanning multiple hours. Each layer consists of a causal 1D convolution with 64 filters and kernel size 5, followed by ReLU activation and dropout regularization (rate = 0.4).

Causal padding ensures that predictions at time t depend only on observations up to t , preserving the temporal structure of forecasting.

The model takes as input a lookback window of 48 consecutive hourly observations across 13 features: lagged $\text{PM}_{2.5}$ values, meteorological variables (wind speed, wind direction), cyclical time encodings (hourly and seasonal sine-cosine pairs), and wildfire proximity measures (three smoke polygon distances and three fire alert distances). The output layer produces a single scalar prediction for $\text{PM}_{2.5}$ concentration one hour ahead.

Input sequences were normalized using min-max scaling fit on the training set. The model was trained using the Adam optimizer with learning rate 10^{-3} and weight decay 10^{-5} . Mean squared error (MSE) served as the loss function. Training proceeded for 5 epochs with batch size 64. The same 80/20 chronological train-test split used for ARIMA models was applied to ensure fair comparison. Model checkpoints were saved based on validation loss, and the best-performing model was retained for evaluation.

5 Results

5.1 ARIMA and SARIMA Baselines

Figure 2 summarizes ARIMA and SARIMA performance across 16 monitoring sites. On average, both models achieve comparable accuracy, with ARIMA slightly outperforming SARIMA in terms of both mean absolute error (MAE) and root mean squared error (RMSE). Incorporating a 24-hour seasonal component does not lead to a substantial improvement in overall performance. At the site level, performance varies across locations. As shown in Figure 2b, some sites benefit modestly from seasonal modeling, while others exhibit lower error under the non-seasonal ARIMA specification, but in general points cluster around the diagonal. Overall, neither model consistently outperforms the other across sites.

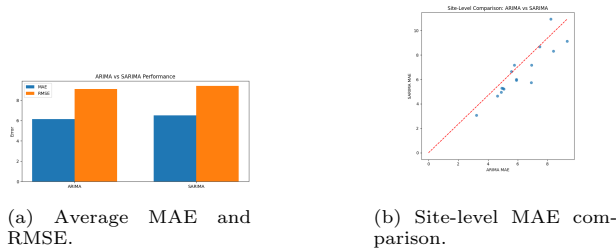


Figure 2: ARIMA vs. SARIMA performance across 16 monitoring sites.

5.2 Model Comparison Across All Sites

Table 1 summarizes performance across 16 sites. TCN achieves the lowest error (MAE: $5.11 \mu\text{g}/\text{m}^3$, RMSE: $11.11 \mu\text{g}/\text{m}^3$), outperforming all baselines. ARIMA performs best among linear models (MAE: $5.37 \mu\text{g}/\text{m}^3$). Adding seasonal components (SARIMA) or exogenous features (SARIMAX) yields marginal improvements. RMSE values remain substantially higher than MAE across all models, indicating persistent large errors during extreme events.

Table 1: Model performance across 16 sites (mean $\pm$ std, all values $\mu\text{g}/\text{m}^3$).

Model	MAE	RMSE
ARIMA	6.13 ± 1.71	9.10 ± 2.68
SARIMA	6.51 ± 2.01	9.41 ± 2.82
TCN	5.11 ± 1.09	11.11 ± 2.27

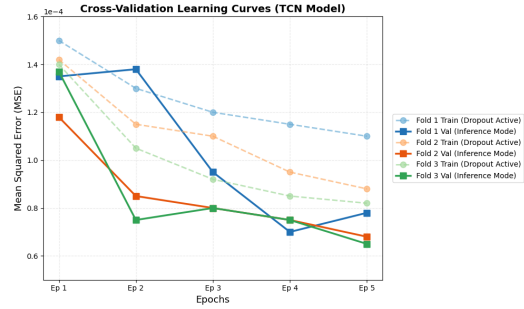


Figure 3: TCN cross-validation learning curves over 5 epochs. Validation MSE (solid lines) remains consistently lower than training MSE (dashed lines) across all three folds. Values shown in scientific notation ($\times 10^{-4}$). This pattern suggests either effective dropout regularization or potential data imbalance between train/test periods.

5.3 Wildfire-Impacted Sites

Table 2 presents metrics for two case studies: Fresno (2020 Creek Fire) and Chico (2021 Dixie Fire). Fresno achieves MAE $5.54 \mu\text{g}/\text{m}^3$ and RMSE $9.89 \mu\text{g}/\text{m}^3$. Chico shows lower MAE ($4.67 \mu\text{g}/\text{m}^3$) but higher RMSE ($12.33 \mu\text{g}/\text{m}^3$), indicating fewer but more severe errors, particularly during an early 2020 event exceeding $380 \mu\text{g}/\text{m}^3$.

During the Dixie Fire period, Chico’s performance degrades substantially: MAE increases 76% to $8.23 \mu\text{g}/\text{m}^3$ and RMSE reaches $15.96 \mu\text{g}/\text{m}^3$. The site recorded 312 hours exceeding EPA’s “Unhealthy” threshold ($35 \mu\text{g}/\text{m}^3$) during this 2,425-hour period.

Table 2: TCN performance at wildfire case study sites (all values $\mu\text{g}/\text{m}^3$).

Metric	Fresno	Chico
Overall MAE	5.54	4.67
Overall RMSE	9.89	12.33
Fire MAE	—	8.23
Fire RMSE	—	15.96
Max Observed	204	386

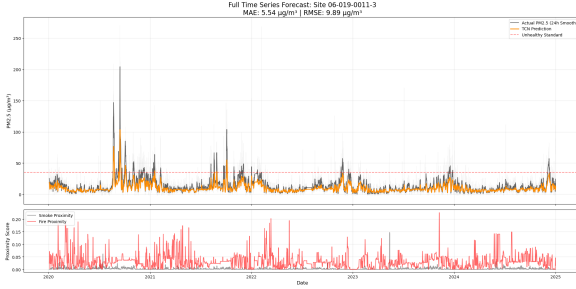


Figure 4: TCN forecast for Fresno (2020-2024). Top: 24h-smoothed $\text{PM}_{2.5}$ (black: actual, orange: predicted). Bottom: wildfire proximity.

5.4 Extreme Event Prediction

During the Dixie Fire, Chico recorded a maximum of $218 \mu\text{g}/\text{m}^3$, but TCN predicted only $80.54 \mu\text{g}/\text{m}^3$ (37% of true peak).

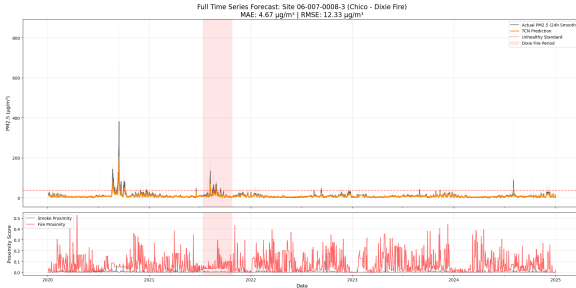


Figure 5: TCN forecast for Chico (2020-2024). Shaded: Dixie Fire (Jul-Oct 2021, 963,309 acres). Top: 24h-smoothed $\text{PM}_{2.5}$. Bottom: wildfire proximity.

5.5 Wildfire Feature Analysis

Bottom panels in Figures 4 and 5 show proximity features. Fire proximity (red) remains elevated during wildfire seasons; smoke proximity (gray) shows sporadic patterns. However, correlation with ground-level $\text{PM}_{2.5}$ is weak. Elevated fire proximity does not consistently predict $\text{PM}_{2.5}$ spikes. The most extreme Chico event (early 2020, $\sim 400 \mu\text{g}/\text{m}^3$) exhibits modest proximity scores, suggesting distance-based features fail to capture atmospheric transport dynamics.

6 Discussion

This study reveals a fundamental tradeoff between average accuracy and extreme event prediction. While TCN achieves lower MAE than ARIMA and SARIMA, it exhibits substantially higher RMSE. This paradox stems from the MSE loss function: quadratic penalties incentivize conservative predictions to avoid massive errors if wrong. During the Dixie Fire, TCN captured only 37% of maximum concentration. This is precisely the wrong failure mode for public health, where missing dangerous events outweighs false alarms. Classical ARIMA models, while less flexible, produce moderate errors in both directions rather than catastrophic underpredictions.

Training dynamics expose another concern: validation loss consistently underperforms training loss. While dropout regularization partially explains this pattern, the magnitude suggests temporal data imbalance between training (2020-mid-2023) and test periods (mid-2023-2024). More critically, extreme events constitute less than 5% of training samples. Minimizing MSE naturally prioritizes abundant background conditions while tolerating large errors on rare extremes where forecasting matters most.

Distance-based wildfire features compound these limitations by failing to capture atmospheric transport physics. Proximity alone cannot represent wind direction effects (upwind fires produce negligible pollution), vertical plume dynamics (whether smoke reaches ground level), or temporal lag (smoke arrives hours after ignition). The 2020 Chico event exemplifies this: $\text{PM}_{2.5}$ exceeded $400 \mu\text{g}/\text{m}^3$ despite modest proximity scores.

Future work should integrate physically informed features (e.g., NOAA HRRR-Smoke operational forecasts, directional wind-weighted smoke exposure) that better represent atmospheric transport dynamics. Additionally, addressing the sparse representation of wildfire features—currently dominated by zeros due to the rarity of fire events—through strategic imputation or feature engineering could improve model learning. Loss functions aligned with public health priorities (weighted MSE, quantile regression) would further help prioritize accuracy during extreme events. These findings demonstrate that model architecture alone cannot overcome inadequate features and training objectives. Reliable extreme event forecasting requires integration of atmospheric physics with data-driven methods.

7 Conclusion

In this study, we examined short-term PM_{2.5} forecasting during wildfire events using classical stationary time-series models and a lightweight deep learning approach. Using hourly PM_{2.5} data from 16 monitoring sites in the Los Angeles region and wildfire information derived from NOAA HMS smoke polygons, we constructed a unified dataset that captures temporal pollution dynamics, nearby wildfire activity, and their spatial relationships.

ARIMA-based models provide interpretable baselines but struggle during abrupt wildfire-driven regime changes. TCNs improve average accuracy by modeling nonlinear temporal structure, yet still underpredict extreme PM_{2.5} events. This suggests that short-term PM_{2.5} forecasting is constrained less by model choice than by feature representation, as proximity-based wildfire indicators offer only a coarse approximation of wildfire influence and fail to capture atmospheric transport and dispersion processes.

Overall, this work demonstrates that lightweight deep learning models can improve short-term PM_{2.5} forecasts relative to classical baselines, but substantial gains will likely require richer representations of wildfire behavior and meteorological dynamics. Future work may benefit from incorporating additional physical and meteorological data sources.

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